

CompTIA Security+ Exam SY0-701

Lesson 6



Secure Cloud Network Architecture

Lesson 6

Topic 6A

Cloud Infrastructure

Cloud Deployment Models

- Public (or multi-tenant)
- Private
 - Hosted Private
- Community
- Hybrid Cloud

Cloud Deployment Models

- Security Considerations
- Single-tenant architecture
- Multi-tenant architecture
- Hybrid architecture
- Serverless architecture

Cloud Service Models

- Models
 - Software as a Service
 - Platform as a Service
 - Infrastructure as a Service
- Third-Party Vendors

Responsibility Matrix

- Describes the balance of responsibility between a customer and a cloud service provider

Responsibility	On-premises	IaaS	PaaS	SaaS	FaaS	CIS Controls Cloud Companion Guide	CIS Foundations Benchmarks
Data classification and accountability	●	●	●	●	●	✓	✓
Client and end-point protection	●	●	●	●	●	✓	✓
Identity and access management	●	●	●	●	●	✓	✓
Application-level controls	●	●	●	●	●	✓	✓
Network controls	●	●	●	●	●	✓	✓
Host infrastructure	●	●	●	●	●	✓	
Physical security	●	●	●	●	●		

● Cloud Customer ● Cloud Provider

Responsibility model

Responsibility Matrix

- Cloud Service Provider
 - Physical security of the infrastructure
 - Securing computer, storage, and network equipment
 - Securing foundational elements of networking, such as DDoS protection
 - Cloud storage backup and recovery
 - Security of cloud infrastructure resource isolation among tenants
 - Tenant resource identity and access control
 - Security, monitoring, and incident response for the infrastructure
 - Securing and managing the datacenters located in multiple geographic regions

Responsibility Matrix

- Cloud Service Customer
 - User identity management
 - Configuring the geographic location for storing data and running services
 - User and service access controls to cloud resources
 - Data and application security configuration
 - Protection of operating systems, when deployed
 - Use and configuration of encryption, especially the protection of keys

Centralized and Decentralized Computing

- Centralized computing architecture
 - All data processing and storage is performed in a single location
 - All users and devices rely on the central server/authority
- Decentralized computing architecture
 - Data processing and storage distributed across multiple locations or devices
 - Increasingly important design trend impacting modern infrastructures

Centralized and Decentralized Computing

- Decentralized computing examples
 - Blockchain
 - Peer-to-peer (P2P)
 - Content Delivery Networks (CDNs)
 - Internet of Things (IoT)
 - Distributed databases
 - TOR (The Onion Router)

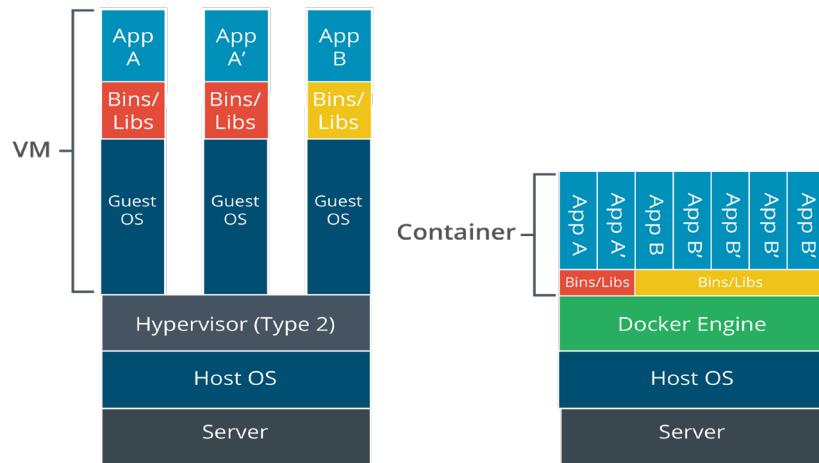
Resilient Architecture Concepts

- Replication
- High Availability Across Zones
 - Local replication
 - Regional replication
 - Geo-redundant storage (GRS)

Application Virtualization and Container Virtualization

- Application virtualization
- Containerization
- Container versus virtual machine

Container vs. VMs



Comparison of VMs versus containers.

Cloud Architecture

- Virtual Private Cloud (VPC)
 - A cloud computing model in which the cloud provider manages the infrastructure and automatically allocates resources as needed, charging only for the actual usage of the application
- Serverless Computing
 - A private network segment made available to a single cloud consumer on a public cloud
- Microservices
 - An architectural approach to building software applications as a collection of small and independent services focusing on a specific business capability

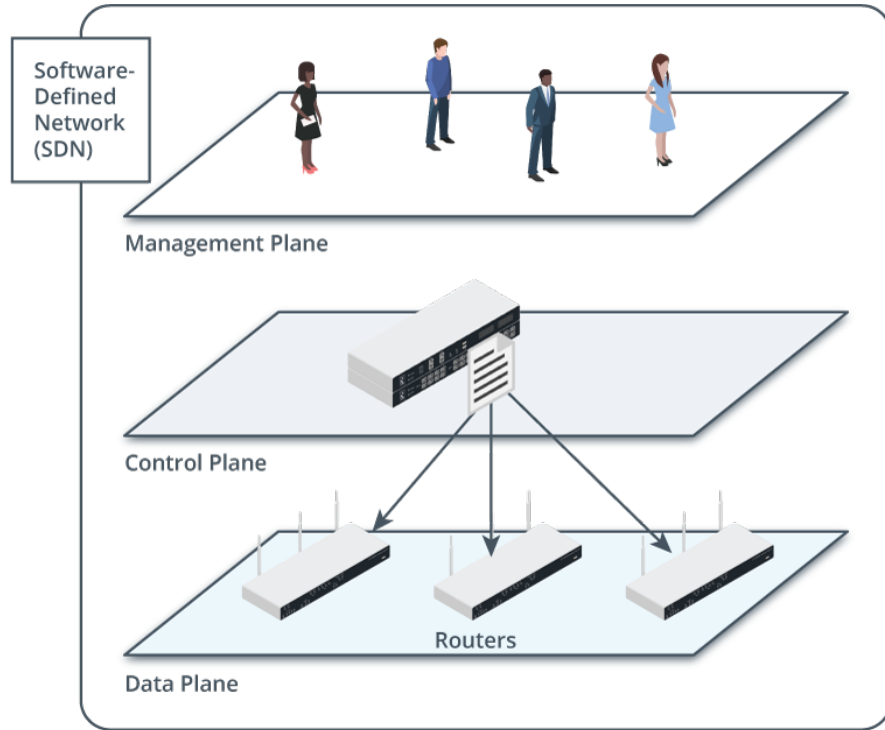
Cloud Automation Technologies

- Infrastructure as Code (IaC)
- Responsiveness
 - Load Balancing
 - Edge Computing
 - Auto-Scaling

Software Defined Networking

- Network functions are divided into three “planes”
- Control plane
 - Decisions about how traffic should be prioritized, secured, and where it should be switched
- Data plane
 - Handles the switching and routing of traffic and imposition of security access controls
- Management plane
 - Monitors traffic conditions and network status

Software Defined Networking



Data plane devices managed by a control plane device and monitored by a management plane. (Images © 123RF.com.)

- SDN is an important part of the latest automation and orchestration technologies
- SDN architecture reduces complexity of enforcing security policy
- Enables fully automated deployment (or provisioning) of network links, appliances, and servers

Cloud Architecture Features

- Considerations for Cloud Infrastructure
- Cost
- Scalability
- Resilience
- Ease of deployment
- Ease of recovery
- SLA and ISA
- Power
- Compute

Cloud Security Considerations

- Considerations for Cloud Infrastructure Security
 - Data protection
 - Patching
 - Secure Communication
 - Software-Defined Wide Area Network (SD-WAN)
 - Secure Access
 - Secure Access Service Edge (SASE)

Review Activity: Cloud Infrastructure

- Cloud Deployment Models
- Cloud Services Models
- Responsibility Matrix
- Centralized and Decentralized Computing
- Resilient Architecture Concepts
- Application Virtualization and Container Virtualization
- Cloud Architecture
- Cloud Automation Technologies
- Software Defined Networking
- Cloud Architecture Features
- Cloud Security Considerations

Lab Activity

- Assisted Lab: Using Containers
- Assisted Lab: Using Virtualization

Lesson 6

Topic 6B

Embedded Systems and Zero Trust Architecture

Embedded Systems

- Specialized computers
- Many consumer and commercial use cases.
- Some examples:
 - Home appliances
 - Smartphones and tablets
 - Automotive systems
 - Industrial automation
 - Medical devices
 - Aerospace and defense
- Real-Time Operating Systems

Industrial Control Systems

- Industrial control systems (ICSs)
 - Human-machine interfaces (HMIs)
 - Data historian
 - Programmable Logic Controller (PLC)
 - Supervisory Control and Data Acquisition (SCADA)
- ICS/SCADA Applications
 - Energy
 - Industrial
 - Fabrication and manufacturing
 - Logistics
 - Facilities

Internet of Things

- Network of physical devices, vehicles, appliances, and other objects embedded with sensors, software, and connectivity, enabling them to collect and exchange data
- The significantly decreased cost of IoT sensors and devices over the past few years has made them more affordable and accessible to businesses and consumers
- Advances in connectivity technology, such as 5G and low-power wireless networks, have made connecting and managing large numbers of IoT devices easier and more efficient

Security Risks for IoT

- Many IoT devices have limited processing power and memory
 - Difficult to implement stringent security controls
- Rushed to market
 - Lacking or misrepresented security capability
 - “Un-patchable”
- Lack of standards in design of IoT devices
- Collect and transmit sensitive information

Best Practice Guidance for IoT

- The Internet of Things Security Foundation (IoTSF)
 - <https://iotsecurityfoundation.org>
- Industrial Internet Consortium (IIC) Security Framework
 - <https://www.iiconsortium.org/iisf/>
- Cloud Security Alliance (CSA) IoT Security Controls Framework
 - <https://cloudsecurityalliance.org/artifacts/iot-security-controls-framework>
- European Telecommunications Standards Institute (ETSI) IoT Security Standards
 - <https://www.etsi.org/technologies/consumer-iot-security>

Deperimeterization and Zero Trust

- Deperimeterization
 - Shifts focus from defending the network boundaries to protecting individual resources
- Zero Trust
 - “Never trust, always verify”

Trends Driving Deperimeterization

- Cloud
- Remote Work
- Mobile
- Outsourcing & Contracting
- Wireless Networks

Key Benefits of a Zero Trust Architecture

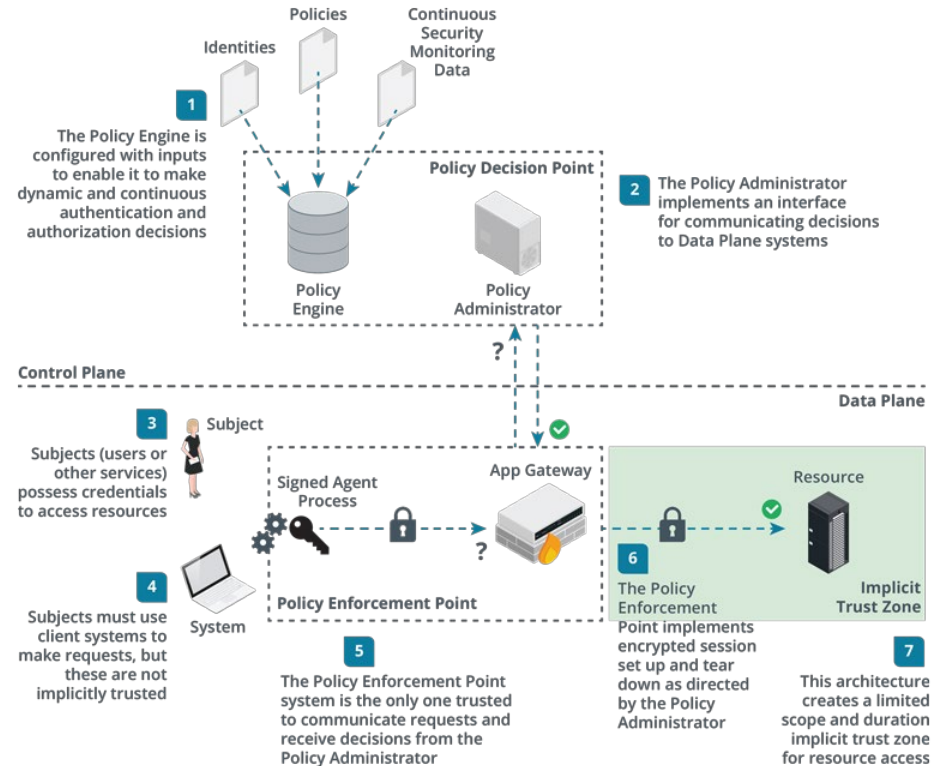
- Greater security
- Better access controls
- Improved governance and compliance
- Increased granularity

Components of a Zero Trust Architecture

- Network and endpoint security
- Identity and access management (IAM)
- Policy-based enforcement
- Cloud security
- Network visibility
- Network segmentation
- Data protection
- Threat detection and prevention

Zero Trust Security Concepts

- Assumes that all devices, users, and services are not inherently trusted, regardless of whether inside or outside a network's perimeter



Components in NIST's zero trust architecture framework.

Review Activity: Embedded Systems and Zero Trust Architecture

- Embedded Systems
- Industrial Control Systems
- Internet of Things
- Deperimeterization and Zero Trust

CompTIA Security+ Exam SY0-701

Lesson 6



Summary